

CSE 4621: Machine Learning

Logistic Regression

Lecture 5

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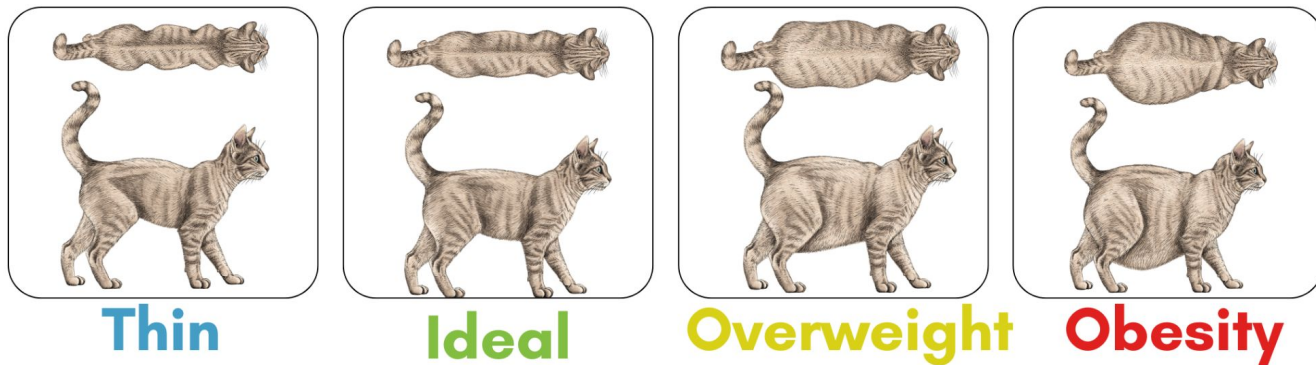
Lecture Outline

- Classification
 - Why not linear regression?
 - New hypothesis function
- Logistic regression
 - Sigmoid function
 - Log-odds and MLE
 - Gradient descent
- Multi-class classification
 - One vs. all

What is Classification?

Classification is a *supervised learning* task where we predict discrete categorical classes

Suppose you want an algorithm that can identify if a cat is obese or not



What is Classification?

We can encode the classes into numbers. But what if there are multiple classes?

Length	Width	Weight	Obese?
15.6	4.5	2.6	Yes
12.5	3.7	1.4	No
...	...	...	...
17.2	5.1	3.1	Yes



Obese?
1
0
...
1

Let's try to fit a simple linear hypothesis to our data, considering only one feature!

What is Classification?

Our hypothesis function:

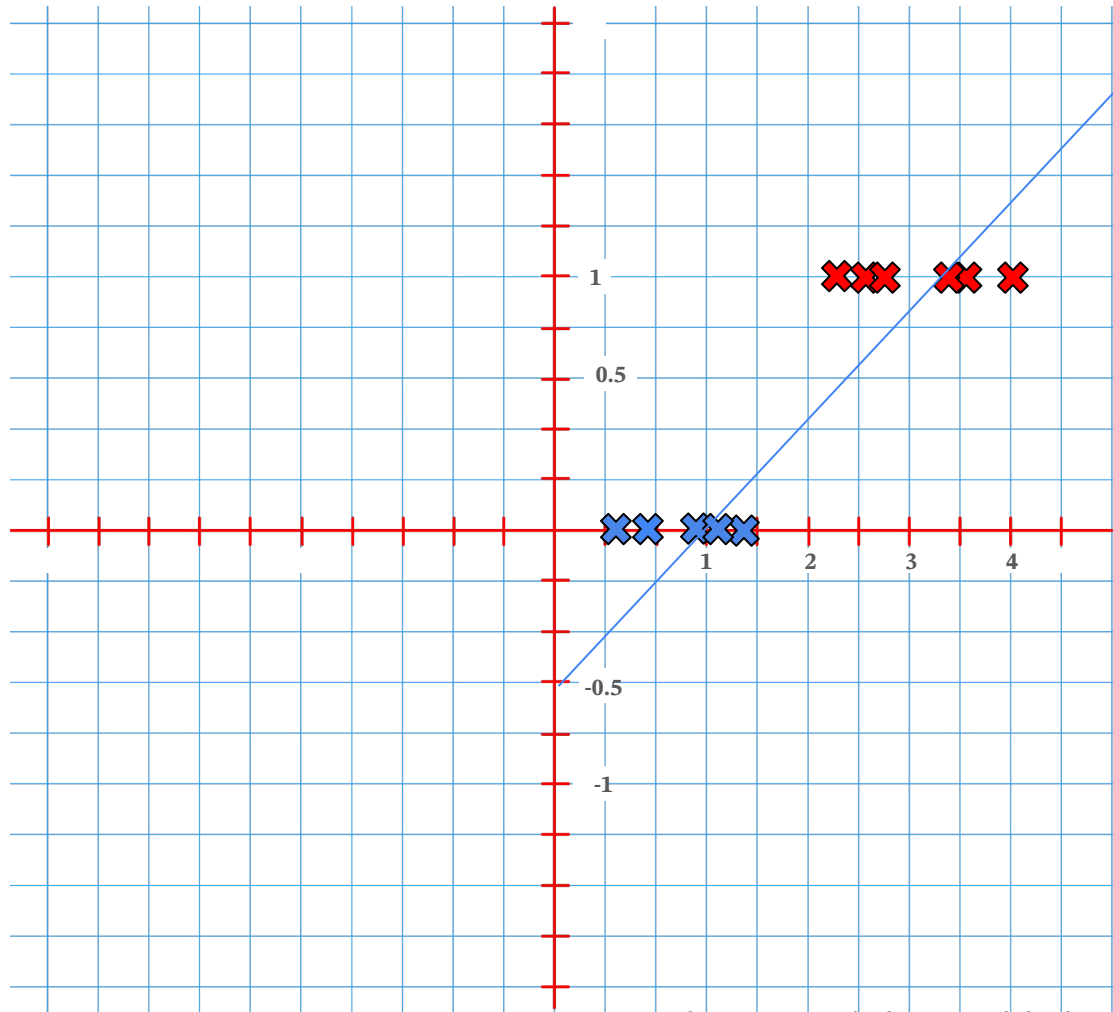
$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 = \theta^T x$$

As our function returns continuous values, to classify different classes, we will use a threshold:

$$h_{\theta}(x) \geq 0.5; y = 1$$

$$h_{\theta}(x) < 0.5; y = 0$$

But, $h_{\theta}(x)$ can be > 1 or < 0 !



What is Classification?

Why can we not just use linear regression?

- The hypothesis function can have values outside of $[0, 1]$
- Predictions are not interpreted as probabilities
- Adding one extreme outlier can totally flip all predictions
- MSE loss is wrong for binary outcomes

We must pose the problem such that our hypothesis function outputs a probability

$h_{\theta}(x) = P(y = 1 | x) \rightarrow$ Probability that $y = 1$ given the input features x

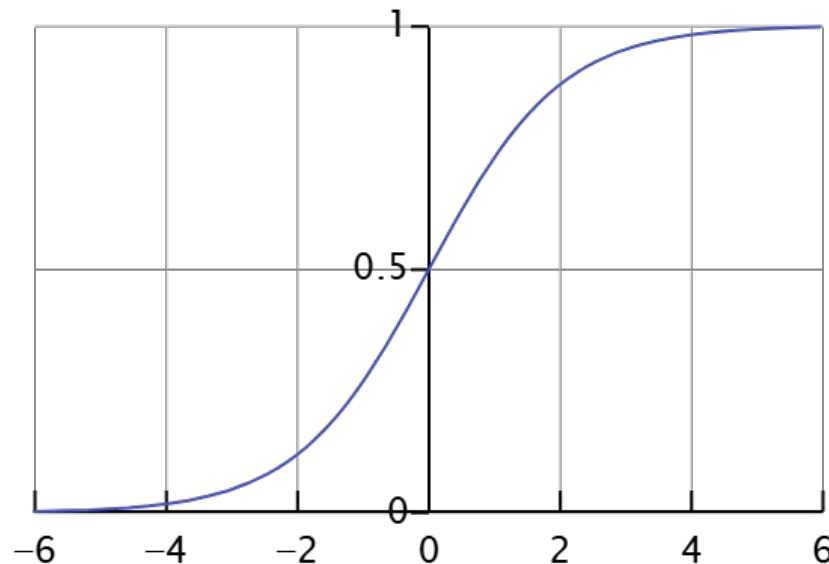
Logistic Regression

We need to rethink our hypothesis function. We want $0 \leq h_{\theta}(x) \leq 1$

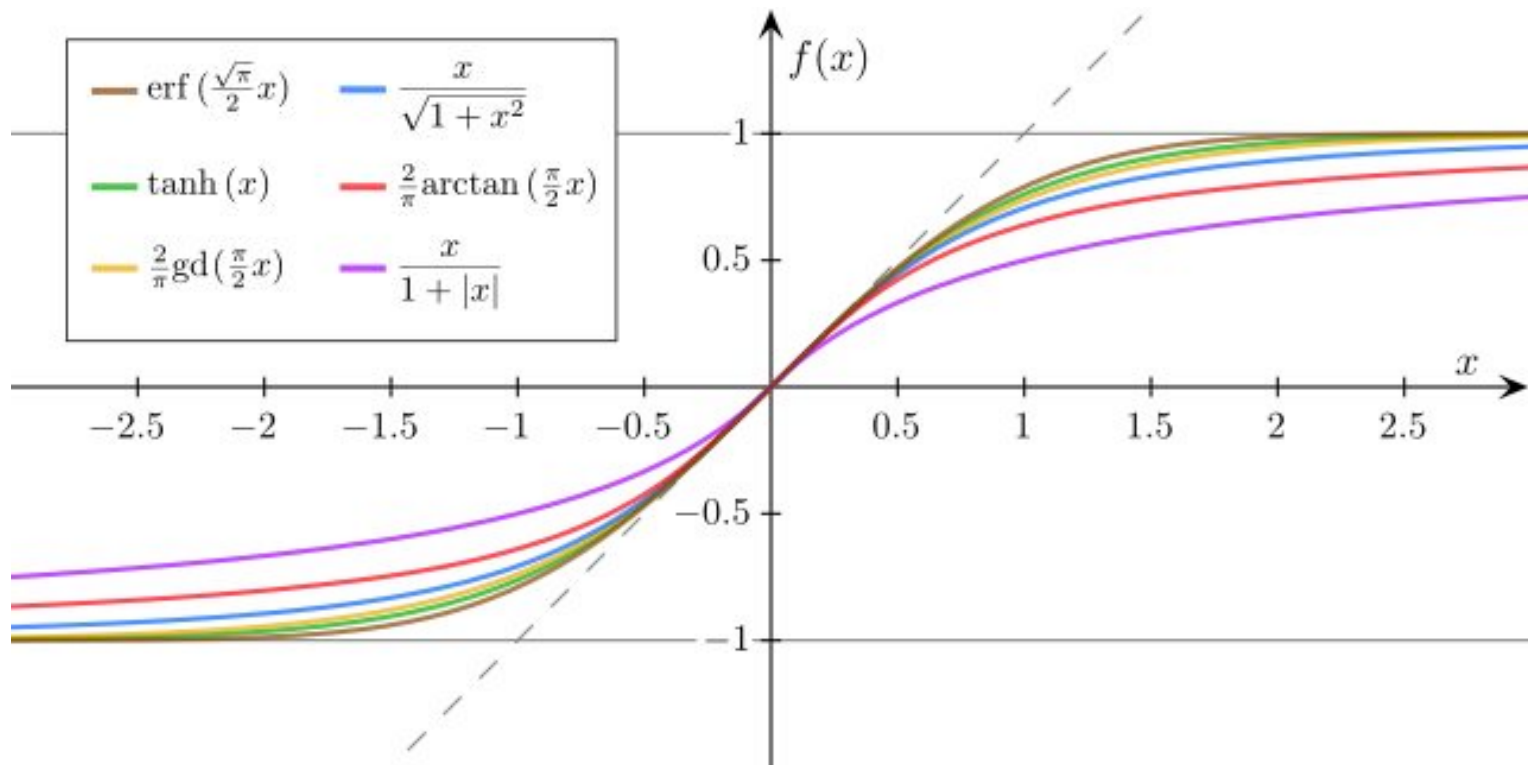
What if we use: $f(x) = \frac{1}{1 + e^{-x}}$

This is called the logistic function

It has a noticeable S-shaped curve, which can be seen in other functions as well, we can call them sigmoid functions



Logistic Regression



Logistic Regression

$$h_{\theta}(x^{(i)}) = \sigma(\theta^T x^{(i)})$$

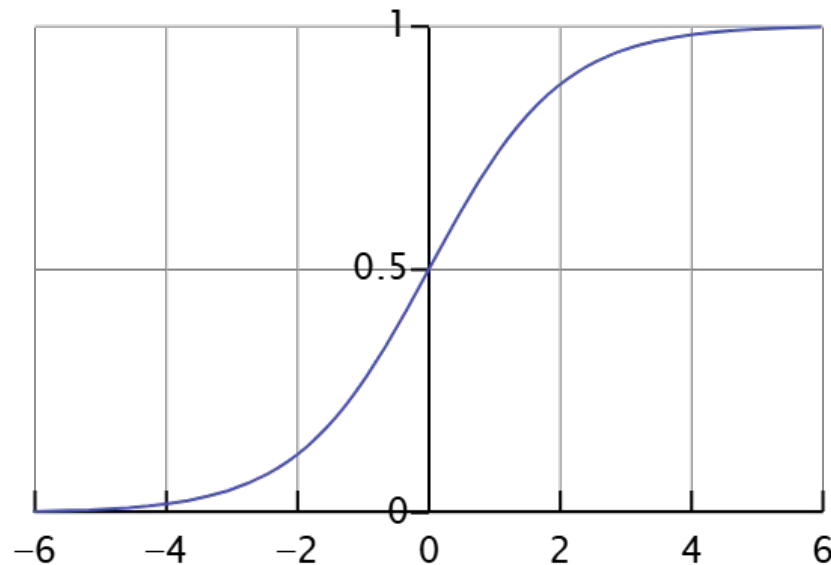
$$\sigma(\theta^T x^{(i)}) = \frac{1}{1 + e^{\theta^T x}}$$

Suppose we set $h_{\theta}(x) \geq 0.5$ to predict $y = 1$

Thus, $\theta^T x \geq 0$

Suppose we set $h_{\theta}(x) < 0.5$ to predict $y = 0$

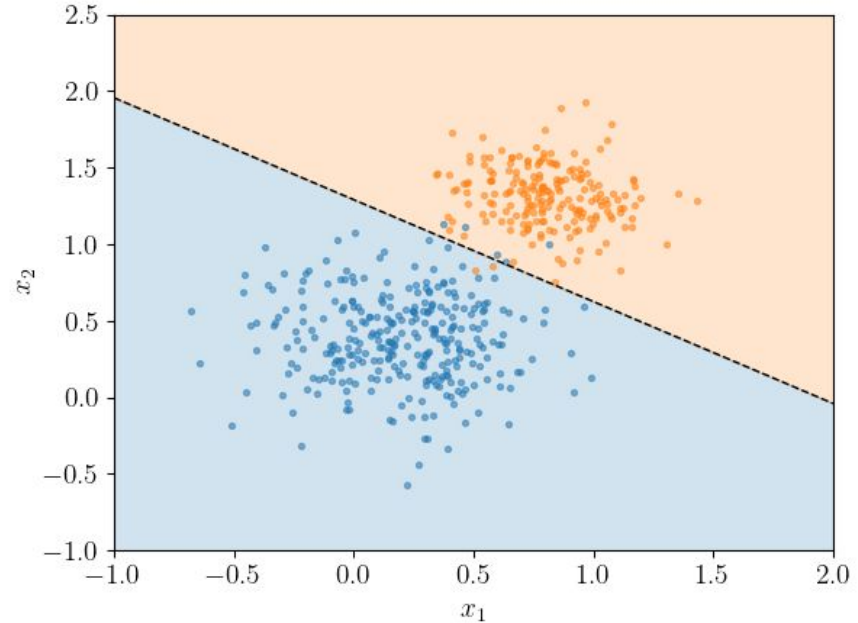
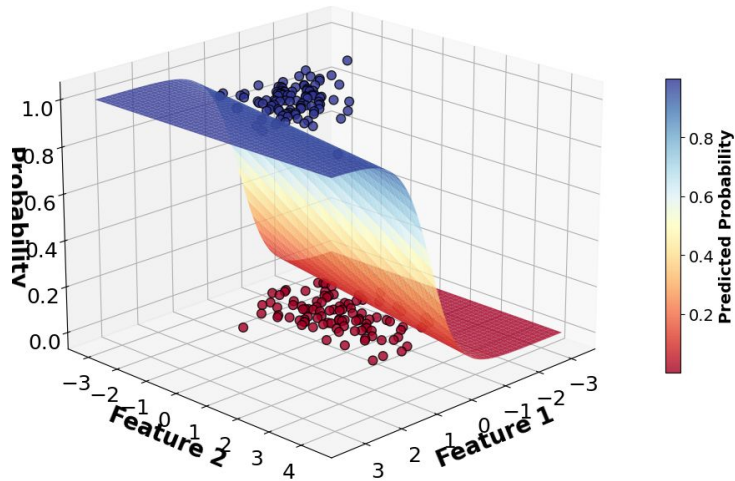
Thus, $\theta^T x < 0$



Logistic Regression

What about multiple features?

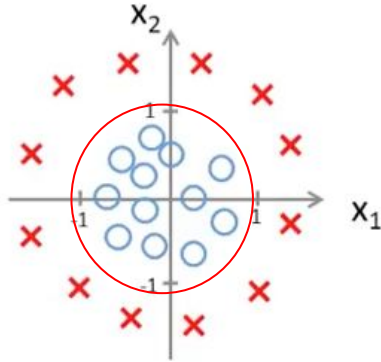
3D Probability Surface



Decision Boundary

The decision boundary is a hyperplane caused by the probability distribution given by the sigmoid function: $\sigma(h_{\theta}(x)) = \sigma(\theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_n x_n)$

For non-linear decision boundaries, use non-linear $h_{\theta}(x)$:

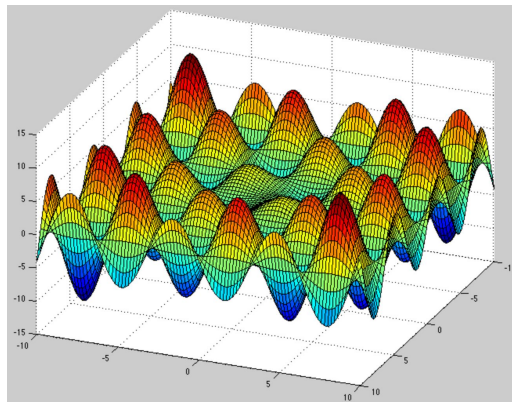


$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2)$$

Cost Function

If we use MSE loss: $J(\theta) = \frac{1}{2m} \sum_{i=1}^m (\sigma(h_{\theta}(x^{(i)})) - y^{(i)})^2$

- This contains a non-linear sigmoid function inside a square
- We are no longer optimizing a simple quadratic function of the parameters θ
- When we square the error after this nonlinear transformation, the curvature of the loss changes sign in different regions.



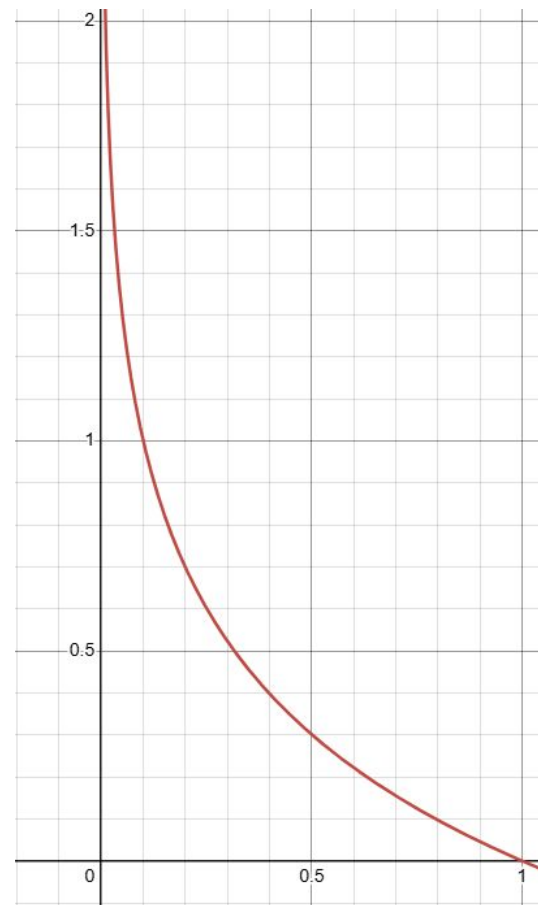
Cost Function

We need to reward *confident correct* predictions, and punish *confident wrong* predictions very heavily.

$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$

If $y=1$ but $h_{\theta}(x)=0$, the cost will be significantly high

If $y=1$ and $h_{\theta}(x)=1$, the cost will be 0



Cost Function

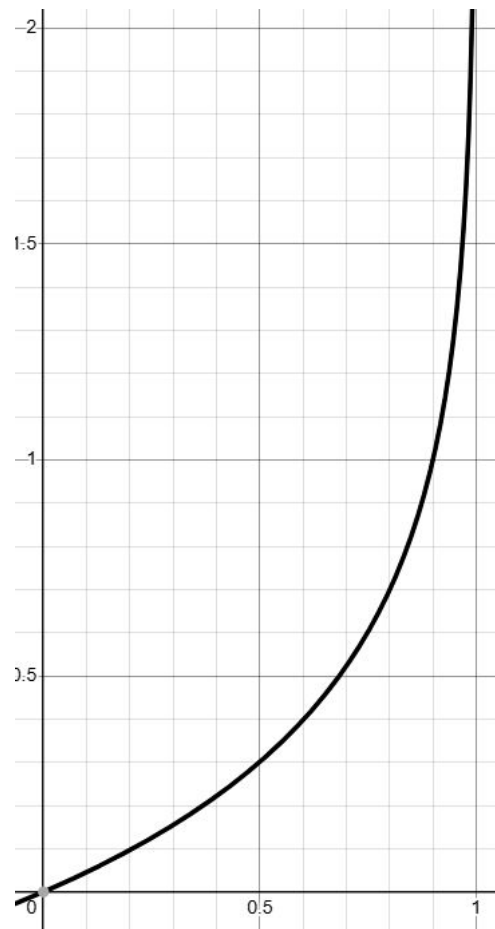
We need to reward *confident correct* predictions, and punish *confident wrong* predictions very heavily.

$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$

If $y=0$ but $h_{\theta}(x)=1$, the cost will be significantly high

If $y=0$ and $h_{\theta}(x)=0$, the cost will be 0

Can we combine these two together?



Cost Function

As the value of y will always be either 0 or 1, we can combine the two cases as:

$$\text{Cost}(h_{\theta}(x), y) = -y \cdot \log(h_{\theta}(x)) - (1 - y) \cdot \log(1 - h_{\theta}(x))$$

$$\text{If } y = 1, \quad \text{Cost}(h_{\theta}(x), y) = -\log(h_{\theta}(x))$$

$$\text{If } y = 0, \quad \text{Cost}(h_{\theta}(x), y) = -\log(1 - h_{\theta}(x))$$

$$\text{Thus, total cost } J(\theta) = \frac{1}{m} \left[\sum_{i=1}^m \text{Cost}(h_{\theta}(x^{(i)}), y^{(i)}) \right]$$

$$J(\theta) = \frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log(h_{\theta}(x^{(i)})) + (1 - y^{(i)}) \log(1 - h_{\theta}(x^{(i)})) \right]$$

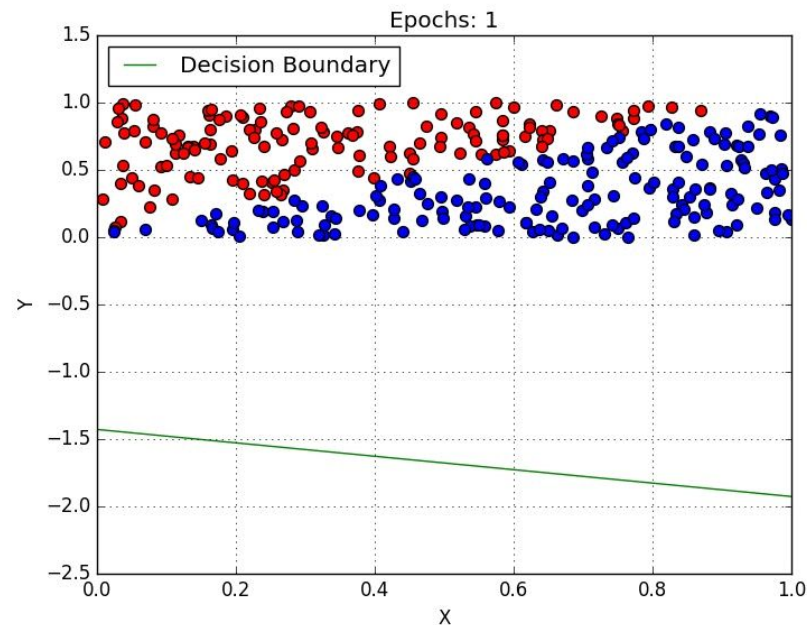
Gradient Descent

Now,

repeat until convergence {

$$\theta_j = \theta_j - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) \cdot x_j^{(i)}$$

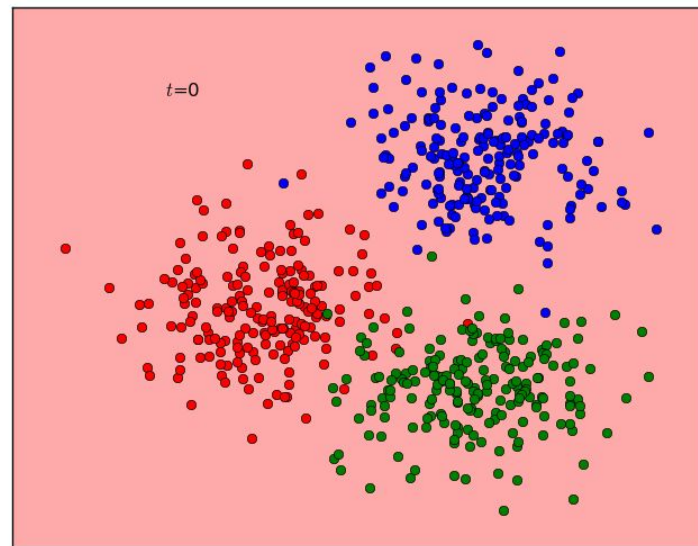
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Multiclass Classification

One-vs-all method:

- Train a classifier for each class i to predict probability that $y = i$
- On a new input x , pick the class that maximizes the probability of being in that class for that x



Textbook Chapters

- [Bishop] - Chapter 4.3.2, 4.3.4

Recommended Resources

1. [Sigmoid Function](#)

Thank you